

The empirical recurrence for OEIS A231705: recovering a conjecture that is not written down

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Abstract

OEIS A231705 records “Empirical recurrence of order 77”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 190$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 77, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 77 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A231705 is “Number of $(n+1) \times (3+1)$ 0..1 arrays with no element having a strict majority of its horizontal, vertical and antidiagonal neighbors equal to one.”. It has offset 1 and begins

42, 387, 2840, 23584, 197605, 1601683, 13087828, 107398293, ...

The entry states:

Empirical recurrence of order 77 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:38:31, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 190$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 190$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 380$ exact terms of the model returns order 77 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{77} c_i a(n-i),$$

c_1	6	c_{21}	10559780	c_{41}	−155465069	c_{61}	−577739
c_2	3	c_{22}	−1659600	c_{42}	−84825093	c_{62}	464139
c_3	149	c_{23}	7876808	c_{43}	22368496	c_{63}	24250
c_4	−258	c_{24}	−3128098	c_{44}	−14272109	c_{64}	−324725
c_5	683	c_{25}	−9342005	c_{45}	−33451466	c_{65}	−133984
c_6	−4012	c_{26}	−25658232	c_{46}	41500769	c_{66}	25853
c_7	9266	c_{27}	−1620210	c_{47}	83775951	c_{67}	21457
c_8	−20216	c_{28}	36891093	c_{48}	−5697125	c_{68}	−14738
c_9	46861	c_{29}	28930178	c_{49}	−39680719	c_{69}	−14093
c_{10}	−120103	c_{30}	−1614581	c_{50}	16130828	c_{70}	−451
c_{11}	297657	c_{31}	28656620	c_{51}	23937260	c_{71}	4492
c_{12}	−349600	c_{32}	−38663639	c_{52}	−26723440	c_{72}	2516
c_{13}	564298	c_{33}	−58020883	c_{53}	−20729737	c_{73}	437
c_{14}	−1987367	c_{34}	−107234952	c_{54}	4732350	c_{74}	−229
c_{15}	2051807	c_{35}	38892846	c_{55}	1241820	c_{75}	−120
c_{16}	−837394	c_{36}	105840081	c_{56}	−4135320	c_{76}	−29
c_{17}	5631724	c_{37}	25258456	c_{57}	5640748	c_{77}	1
c_{18}	−7427655	c_{38}	60154632	c_{58}	7630123		
c_{19}	147034	c_{39}	85067273	c_{59}	960445		
c_{20}	−5044942	c_{40}	−8760283	c_{60}	−2323869		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 77, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $77 + 4$ terms removed returns order 77 as well, so $\deg q_{\min} = 77 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $77 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 77 that this sequence satisfies — and that family turns out to have exactly one member.

References

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